DANIEL MORA

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EDUCATION

IE University **Madrid, Spain**
Bachelor in Computer Science and Artificial Intelligence 2026

EXPERIENCE

WhiteHole (hedge fund) **San Sebastián, Spain**
Quantitative Research Intern Jun 2025 – Nov 2025

- Forecast CPI and GDP from housing-market indicators, from feature engineering to out-of-sample validation.
- Engineered NLP features from public political communications and tested their power to predict market volatility with tree-based, linear and ridge regression models, including regime shifts.
- Built a knowledge graph from text using NLP extraction and locally hosted language models.

MAI-Book **Remote**
Front-end Developer Intern Jun 2023 – Oct 2023

- Built user-facing interfaces for an online library; ran A/B tests and UX research to lift engagement and retention.

RESEARCH AND PROJECTS

JEPA-Embedding Transformer (ArgusJEPA + CassaFormer) PyTorch, Triton 2026

- Causal Transformer forecasting next-day US equity targets from JEPA embeddings; code and weights public.
- ArgusJEPA (self-supervised, EMA teacher, 50% masking) embeds each stock-day in 256-d; frozen, these feed CassaFormer, a 2.05M-parameter decoder with RoPE, QK-RMSNorm, top-1 MoE and a Student-t head.
- Point-in-time data (IBKR, SEC, FRED/ALFRED, news) on PostgreSQL + DuckDB; survivorship-free top 500.
- Wrote a custom Triton GPU kernel for the MoE layers: up to 1.5× end-to-end speedup.

TC_ACC: Agentic Documentary Studio Python, LangGraph, Remotion, ffmpeg 2026

- Agentic harness that turns one case reference into a finished 10-minute documentary in about five hours with no human in the loop: it researches, writes, narrates, selects stock footage and edits the final cut.
- Published episodes have drawn 400+ watch hours on YouTube.
- 28 stages, 5 departments, 2 review gates; typed Pydantic artifacts, resumable stages, 46 keyless CI tests.

Brain–LLM Sparse Feature Correspondence (BA Thesis) PyTorch, SAEs, fMRI, ridge, RSA 2026

- Tested whether brain and LM mappings share sparse features: 59 fMRI participants, six SAE systems (1B–12B).
- A frozen SAE feature basis predicted held-out brain responses ($r \approx 0.07$–0.09), but brain and model feature-importance profiles did not align ($\rho = -0.06$ vs. temporal nulls).

Sparse Autoencoder Releases PyTorch, Hugging Face, GCP

- OLMo-3 SAEs: trained and released seven TopK sparse autoencoders across four layers of OLMo-3-7B-Think.
- Triplet extractor: SAE steering vs. LoRA; the fine-tuned Gemma 3 4B gets $\sim$50 monthly downloads.

LEADERSHIP

- **Founder and President**, IE Dominican Republic Club: IE's largest student-led diversity event (500+).
- **Founder**, IE Quant Finance Club; **Business Development Officer**, IE Investment Club.

SKILLS

ML: PyTorch, Triton, Hugging Face, JEPA, sparse autoencoders, state-of-the-art LLM architectures
Modeling: Time-series forecasting, linear/ridge/tree models, scikit-learn, Granger causality, regime analysis
Agents: LangGraph, Pydantic, multi-stage agent workflows, RAG, GraphRAG, local LLM inference
Systems: Python, C++, SQL, PostgreSQL, DuckDB, Redis, FastAPI, Docker, GitHub Actions, GCP, Azure
Languages: Spanish (native), English (fluent)